A CHAIN OF MORPHISMS OF COMPLEXES FACTORS THROUGH TERMWISE SPLIT INJECTIONS UP TO HOMOTOPY EQUIVALENCE

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Editorial note. Counted span is the author's own proof environment, source lines 3381-3390, immediately following the statement at 3356-3379.

Editorial note. The boundedness clause of the statement is carried by the same induction; the source text does not spell it out.

Allowed standard prerequisites (editorial).

- The two-object case. Lemma lemma-make-injective of this chapter, cited at proof lines 3383-3384 and 3386-3390. Supplies the n = 2 case and the inductive step: any morphism factors B -> B' -> A' with B -> B' termwise split injection and B' -> A' homotopy equivalence.

Source reference. Original lemma 013N: sources/stacks-derived/pinned-original-source.tex, source lines 2840–2840. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an additive category. Let $A_1^\bullet \to A_2^\bullet \to \ldots \to A_n^\bullet$ be a sequence of composable morphisms of complexes. There exists a commutative diagram*

$$\begin{array}{ccccccc} A_1^\bullet & \longrightarrow & A_2^\bullet & \longrightarrow & \ldots & \longrightarrow & A_n^\bullet \\ \uparrow & & \uparrow & & & & \uparrow \\ B_1^\bullet & \longrightarrow & B_2^\bullet & \longrightarrow & \ldots & \longrightarrow & B_n^\bullet \end{array}$$

such that each morphism $B_i^\bullet \to B_{i+1}^\bullet$ is a termwise split injection and each $B_i^\bullet \to A_i^\bullet$ is a homotopy equivalence. Moreover, if all $A_i^\bullet$ are in $K^+(\mathcal{A})$, $K^-(\mathcal{A})$, or $K^b(\mathcal{A})$, then so are the $B_i^\bullet$.

Proof. The case $n = 1$ is without content. Lemma 013N is the case $n = 2$. Suppose we have constructed the diagram except for $B_n^\bullet$. Apply Lemma 013N to the composition $B_{n-1}^\bullet \to A_{n-1}^\bullet \to A_n^\bullet$. The result is a factorization $B_{n-1}^\bullet \to B_n^\bullet \to A_n^\bullet$ as desired. □

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